First Non-Repeating Character

Week 1 Practice - Problem 4 | Java

Problem Summary

Count the frequency of each character, then scan left-to-right and return the first one that occurs exactly once. If none exists, return null.

Java Solution

```
import java.util.LinkedHashMap;
import java.util.Map;
public class Main {
    static Character findFirstNonRepeatingChar(String text) {
        Map<Character, Integer> frequency = new LinkedHashMap<>();
        for (int i = 0; i < text.length(); i++) {
            char c = text.charAt(i);
            frequency.put(c, frequency.getOrDefault(c, 0) + 1);
        }
        for (int i = 0; i < text.length(); i++) {
            char c = text.charAt(i);
            if (frequency.get(c) == 1) return c;
        }
        return null;
    }
    static void printResult(String text) {
        Character c = findFirstNonRepeatingChar(text);
        if (c == null) System.out.println("No Non-Repeating Character Found");
        else System.out.println("First Non-Repeating Character: '" + c + "'");
    }
    public static void main(String[] args) {
        printResult("swiss"); printResult("aabbcc");
    }
}
```

Sample Output

```
First Non-Repeating Character: 'w'
No Non-Repeating Character Found
```

Complexity / Key Notes

Average time: O(n). Extra space: O(k) for k distinct characters. The source permits a hash map or array-based counting; this solution uses LinkedHashMap.